URBAN AND CRITICAL INFRASTRUCTURE RESILIENCE

URBAN RISK SYSTEM → CRITICALITY TEST → INTERDEPENDENCY WEB → RESILIENCE TRIAD → CONTINUITY PLAN → URBAN PLANNING STACK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure: the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◈ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

00

STAGE 00 URBAN RISK SYSTEM

URBAN RISK SYSTEM · HAZARD + DENSITY + EXPOSED ASSETS + UNEQUAL VULNERABILITY · INTERDEPENDENT WATER POWER TRANSPORT HEALTH TELECOM · CASCADING SERVICE FAILURE · SYSTEMIC URBAN DISASTER

HAZARD + DENSITY + EXPOSED ASSETS + UNEQUAL VULNERABILITY

INTERDEPENDENT WATER POWER TRANSPORT HEALTH TELECOM → CASCADING SERVICE FAILURE → SYSTEMIC URBAN DISASTER

HAZARD / RISK / RESILIENCE CONCEPT
- HAZARD + DENSITY + EXPOSED ASSETS + UNEQUAL VULNERABILITY
- INTERDEPENDENT WATER POWER TRANSPORT HEALTH TELECOM

INSTITUTIONAL OR COMMUNITY RESPONSE
- CASCADING SERVICE FAILURE
- SYSTEMIC URBAN DISASTER

CAPACITY / STATUS / EVIDENCE LIMIT
- HAZARD + DENSITY + EXPOSED ASSETS + UNEQUAL VULNERABILITY
- INTERDEPENDENT WATER POWER TRANSPORT HEALTH TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HAZARD + DENSITY + EXPOSED ASSETS + UNEQUAL VULNERABILITY.

01

STAGE 01 CRITICALITY TEST

CRITICALITY TEST · IF FAILURE · DISPROPORTIONATE LIFE · GOVERNANCE EFFECT · THEN CRITICAL · LIFELINE FUNCTION · PRIORITY DEPENDS ON CONSEQUENCE, NOT OWNERSHIP OR SIZE

ASSET / NETWORK / SERVICE

IF FAILURE → DISPROPORTIONATE LIFE / HEALTH / GOVERNANCE EFFECT → THEN CRITICAL / LIFELINE FUNCTION → PRIORITY DEPENDS ON CONSEQUENCE, NOT OWNERSHIP OR SIZE

HAZARD / RISK / RESILIENCE CONCEPT
- ASSET / NETWORK / SERVICE
- IF FAILURE → DISPROPORTIONATE LIFE / HEALTH / GOVERNANCE EFFECT

INSTITUTIONAL OR COMMUNITY RESPONSE
- THEN CRITICAL / LIFELINE FUNCTION
- PRIORITY DEPENDS ON CONSEQUENCE, NOT OWNERSHIP OR SIZE

CAPACITY / STATUS / EVIDENCE LIMIT
- ASSET / NETWORK / SERVICE
- IF FAILURE → DISPROPORTIONATE LIFE / HEALTH / GOVERNANCE EFFECT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRIORITY DEPENDS ON CONSEQUENCE, NOT OWNERSHIP OR SIZE.

02

STAGE 02 INTERDEPENDENCY WEB

INTERDEPENDENCY WEB · WATER PUMPS · REPAIR ACCESS · SHARED NODE FAILURE · MULTI-SYSTEM CASCADE

POWER → WATER PUMPS / HOSPITALS / TELECOM

TELECOM → WARNING / DISPATCH / PAYMENT → TRANSPORT → STAFF / SUPPLY / REPAIR ACCESS → SHARED NODE FAILURE → MULTI-SYSTEM CASCADE

HAZARD / RISK / RESILIENCE CONCEPT
- POWER → WATER PUMPS / HOSPITALS / TELECOM
- TELECOM → WARNING / DISPATCH / PAYMENT

INSTITUTIONAL OR COMMUNITY RESPONSE
- TRANSPORT → STAFF / SUPPLY / REPAIR ACCESS
- SHARED NODE FAILURE → MULTI-SYSTEM CASCADE

CAPACITY / STATUS / EVIDENCE LIMIT
- POWER → WATER PUMPS / HOSPITALS / TELECOM
- TELECOM → WARNING / DISPATCH / PAYMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → STAFF / SUPPLY / REPAIR ACCESS.

03

STAGE 03 RESILIENCE TRIAD

RESILIENCE TRIAD · ALTERNATE CAPACITY · RAPID RECOVERY · RESTORE PRIORITY FUNCTION · COMPLETE RESILIENCE NEEDS ALL THREE

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
ROBUSTNESS → RESIST	RAPID RECOVERY → RESTORE PRIORITY FUNCTION	ROBUSTNESS → RESIST
REDUNDANCY / DIVERSITY → ALTERNATE CAPACITY	COMPLETE RESILIENCE NEEDS ALL THREE	REDUNDANCY / DIVERSITY → ALTERNATE CAPACITY

HAZARD / ACTOR / INSTRUMENT
- ROBUSTNESS → RESIST
- REDUNDANCY / DIVERSITY → ALTERNATE CAPACITY

DECISIVE RISK OR MANDATE DISTINCTION
- RAPID RECOVERY → RESTORE PRIORITY FUNCTION
- COMPLETE RESILIENCE NEEDS ALL THREE

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- ROBUSTNESS → RESIST
- REDUNDANCY / DIVERSITY → ALTERNATE CAPACITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAPID RECOVERY → RESTORE PRIORITY FUNCTION; COMPLETE RESILIENCE NEEDS ALL THREE.

04

STAGE 04 CONTINUITY PLAN

CONTINUITY PLAN · DEFINE MINIMUM SERVICE + PRIORITY USERS · MAP STAFF SUPPLY DATA SITE AND UTILITY DEPENDENCIES · PROVIDE BACKUP · MANUAL MODE · SET RESTORATION ORDER TIME AND COMMUNICATION

DEFINE MINIMUM SERVICE + PRIORITY USERS → MAP STAFF SUPPLY DATA SITE AND UTILITY DEPENDENCIES → PROVIDE BACKUP / ALTERNATE / MANUAL MODE → SET RESTORATION ORDER TIME AND COMMUNICATION

RISK CONDITION / INPUT
- DEFINE MINIMUM SERVICE + PRIORITY USERS
- MAP STAFF SUPPLY DATA SITE AND UTILITY DEPENDENCIES

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- PROVIDE BACKUP / ALTERNATE / MANUAL MODE
- SET RESTORATION ORDER TIME AND COMMUNICATION

LAST-MILE / INCLUSION / OUTCOME LIMIT
- DEFINE MINIMUM SERVICE + PRIORITY USERS
- MAP STAFF SUPPLY DATA SITE AND UTILITY DEPENDENCIES

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP STAFF SUPPLY DATA SITE AND UTILITY DEPENDENCIES.

05

STAGE 05 URBAN PLANNING STACK

URBAN PLANNING STACK · LAND USE · DRAINAGE PATH · BUILDING AND DEVELOPMENT CONTROL · WATER BODY DRAIN SOLID-WASTE AND MAINTENANCE · WARD-CITY-CATCHMENT COORDINATION

LAND USE / FLOODPLAIN / DRAINAGE PATH

BUILDING AND DEVELOPMENT CONTROL → WATER BODY DRAIN SOLID-WASTE AND MAINTENANCE → WARD-CITY-CATCHMENT COORDINATION

HAZARD / RISK / RESILIENCE CONCEPT
- LAND USE / FLOODPLAIN / DRAINAGE PATH
- BUILDING AND DEVELOPMENT CONTROL

INSTITUTIONAL OR COMMUNITY RESPONSE
- WATER BODY DRAIN SOLID-WASTE AND MAINTENANCE
- WARD-CITY-CATCHMENT COORDINATION

CAPACITY / STATUS / EVIDENCE LIMIT
- LAND USE / FLOODPLAIN / DRAINAGE PATH
- BUILDING AND DEVELOPMENT CONTROL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER BODY DRAIN SOLID-WASTE AND MAINTENANCE; WARD-CITY-CATCHMENT COORDINATION.

06

STAGE 06 RESILIENCE AUDIT

RESILIENCE AUDIT · 1 CRITICAL FUNCTION AND USERS · 2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS · 3 BACKUP DURATION + RECOVERY TIME · 4 PRIORITISED RETROFIT CONTINUITY AND DRILL ACTION

1 CRITICAL FUNCTION AND USERS

2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS → 3 BACKUP DURATION + RECOVERY TIME → 4 PRIORITISED RETROFIT CONTINUITY AND DRILL ACTION

RISK CONDITION / INPUT
- 1 CRITICAL FUNCTION AND USERS
- 2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- 3 BACKUP DURATION + RECOVERY TIME
- 4 PRIORITISED RETROFIT CONTINUITY AND DRILL ACTION

LAST-MILE / INCLUSION / OUTCOME LIMIT
- 1 CRITICAL FUNCTION AND USERS
- 2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2 HAZARD LOAD + DEPENDENCIES + SINGLE POINTS.

07

STAGE 07 ESSENTIAL FACILITY SHELL

ESSENTIAL FACILITY SHELL · FLOOD SAFETY · POWER WATER CLINICAL SUPPLY AND TELECOM · ACCESS STAFF REFERRAL AND SECURITY · CONTINUE CARE WHILE DAMAGE IS REPAIRED

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
STRUCTURE / FIRE / FLOOD SAFETY	ACCESS STAFF REFERRAL AND SECURITY	STRUCTURE / FIRE / FLOOD SAFETY
POWER WATER CLINICAL SUPPLY AND TELECOM	CONTINUE CARE WHILE DAMAGE IS REPAIRED	POWER WATER CLINICAL SUPPLY AND TELECOM

HAZARD / ACTOR / INSTRUMENT
- STRUCTURE / FIRE / FLOOD SAFETY
- POWER WATER CLINICAL SUPPLY AND TELECOM

DECISIVE RISK OR MANDATE DISTINCTION
- ACCESS STAFF REFERRAL AND SECURITY
- CONTINUE CARE WHILE DAMAGE IS REPAIRED

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- STRUCTURE / FIRE / FLOOD SAFETY
- POWER WATER CLINICAL SUPPLY AND TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACCESS STAFF REFERRAL AND SECURITY; CONTINUE CARE WHILE DAMAGE IS REPAIRED.

08

STAGE 08 EQUITY ACCESS TEST

EQUITY ACCESS TEST · WHO LIVES IN HIGHEST EXPOSURE? · WHO DEPENDS ON INTERRUPTIBLE OR INFORMAL SERVICES? · WHO CAN REACH WARNING SHELTER HEALTH AND TRANSPORT? · RISK REDUCTION WITHOUT EXCLUSIONARY DISPLACEMENT

HAZARD / RISK / RESILIENCE CONCEPT
- WHO LIVES IN HIGHEST EXPOSURE?
- WHO DEPENDS ON INTERRUPTIBLE OR INFORMAL SERVICES?

INSTITUTIONAL OR COMMUNITY RESPONSE
- WHO CAN REACH WARNING SHELTER HEALTH AND TRANSPORT?
- RISK REDUCTION WITHOUT EXCLUSIONARY DISPLACEMENT

CAPACITY / STATUS / EVIDENCE LIMIT
- WHO LIVES IN HIGHEST EXPOSURE?
- WHO DEPENDS ON INTERRUPTIBLE OR INFORMAL SERVICES?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WHO CAN REACH WARNING SHELTER HEALTH AND TRANSPORT?.

09

STAGE 09 GOVERNANCE MAP

GOVERNANCE MAP · DEVELOPMENT AUTHORITY · SECTOR REGULATOR + SDMA · COORDINATION DOES NOT ERASE OWNER DUTY · ENABLING LAW · MUST REMEMBER · URBAN AND CRITICAL-INFRASTRUCTURE RESILIENCE MAPS INTERDEPENDENT POWER

ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT

SECTOR REGULATOR + SDMA / NDMA → COORDINATION DOES NOT ERASE OWNER DUTY → RECOMMENDATION / ENABLING LAW / NOTIFICATION / OPERATION

HAZARD / RISK / RESILIENCE CONCEPT
- ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT
- SECTOR REGULATOR + SDMA / NDMA

INSTITUTIONAL OR COMMUNITY RESPONSE
- COORDINATION DOES NOT ERASE OWNER DUTY
- RECOMMENDATION / ENABLING LAW / NOTIFICATION / OPERATION

CAPACITY / STATUS / EVIDENCE LIMIT
- MUST REMEMBER: Urban and critical-infrastructure resilience maps interdependent power,...
- ULB / UTILITY / DEVELOPMENT AUTHORITY / DISTRICT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Urban and critical-infrastructure resilience maps interdependent power,.

10

STAGE 10 SMART RESILIENCE FIREWALL

SMART RESILIENCE FIREWALL · DIGITAL TWIN · COMMAND PLATFORM · PRIVACY + CYBERSECURITY + INTEROPERABILITY · OFFLINE FALLBACK + HUMAN ACCOUNTABILITY · EFFICIENCY WITHOUT BUFFER · BRITTLE SYSTEM · CLOSE DISTINCTION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS	OFFLINE FALLBACK + HUMAN ACCOUNTABILITY	CLOSE DISTINCTION: Critical is not merely expensive; redundancy is not unused...
PRIVACY + CYBERSECURITY + INTEROPERABILITY	EFFICIENCY WITHOUT BUFFER → BRITTLE SYSTEM	SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS

HAZARD / ACTOR / INSTRUMENT
- SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS
- PRIVACY + CYBERSECURITY + INTEROPERABILITY

DECISIVE RISK OR MANDATE DISTINCTION
- OFFLINE FALLBACK + HUMAN ACCOUNTABILITY
- EFFICIENCY WITHOUT BUFFER → BRITTLE SYSTEM

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- CLOSE DISTINCTION: Critical is not merely expensive; redundancy is not unused...
- SENSORS / DIGITAL TWIN / COMMAND PLATFORM → AWARENESS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Critical is not merely expensive; redundancy is not unused.

11

STAGE 11 URBAN RESILIENCE ANSWER SPINE

URBAN RESILIENCE ANSWER SPINE · DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES · MAP CASCADES · ROBUSTNESS REDUNDANCY CONTINUITY · ADD LAND USE CODES DRAINAGE AUDIT EQUITY AND GOVERNANCE · USE CDRI · TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS · EVIDENCE LIMIT

DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES → MAP CASCADES → ROBUSTNESS REDUNDANCY CONTINUITY → ADD LAND USE CODES DRAINAGE AUDIT EQUITY AND GOVERNANCE → USE CDRI / TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS → EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name service level, dependency

HAZARD / RISK / RESILIENCE CONCEPT
- DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES
- MAP CASCADES → ROBUSTNESS REDUNDANCY CONTINUITY

INSTITUTIONAL OR COMMUNITY RESPONSE
- ADD LAND USE CODES DRAINAGE AUDIT EQUITY AND GOVERNANCE
- USE CDRI / TECHNOLOGY WITH STATUS AND OUTCOME FIREWALLS

CAPACITY / STATUS / EVIDENCE LIMIT
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name service level, dependency,...
- DEFINE SYSTEMIC URBAN RISK AND CRITICAL LIFELINES

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name service level, dependency,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name service level, dependency..

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · CDRI'S PRECISE INSTITUTIONAL PURPOSE · TO DEVELOP "COMMON · URBAN DISASTER MANAGEMENT AUTHORITY STATUTORY STATUS · FIVE-CITY URBAN-FLOOD EVIDENCE BASE AS A CRITICAL-INFRASTRUCTURE · CDRI'S CATASTROPHE-BOND AND RESERVE-FUND TOOLS AS UNDERUSED IN · CRITICALITY MAPPING IS IMPLIED BUT NOT NAMED AS A · LIFELINE-FAILURE IMPACT IS NOT EVENLY DISTRIBUTED

OPTIONAL RISK / RESILIENCE DEPTH
- CDRI's precise institutional purpose ; to develop "common
- Urban Disaster Management Authority statutory status : the
- Five-city urban-flood evidence base as a critical-infrastructure
- CAUTION: CDRI's catastrophe-bond and reserve-fund tools as underused in

MANDATE, WARNING OR OUTCOME LIMIT
- CAUTION: Criticality mapping is implied but not named as a distinct
- CAUTION: Lifeline-failure impact is not evenly distributed: informal
- CAUTION: Retrofitting existing lifeline infrastructure versus new-build
- CAUTION: "Build Back Better" for management systems versus physical

COMPARATIVE HAZARD / GOVERNANCE NUANCE
- NOT: The "CDRI membership/Urban Flooding Cell exists, therefore urban
- CAUTION: International-standard-setting versus domestic-operational
- CDRI's financing thematic area names budget reserve funds and
- Any current claim about Indian cities' actual use of catastrophe

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels